

SC4172 Tiny Machine Machine Learning

Tutorial 4 – CNN Convolution

In this tutorial, we will analysis the complexity of a conventional CNN (which we will compare against a Depthwise Separable Convolution network in a upcoming tutorial).

1. A 12x12 color image is to be convoluted with a CNN layer that consists of 256 5x5 kernels, with no padding and a stride of 1.
 - a. What is the resultant size of the output tensor?
 - b. Calculate the total number of MAC operations.
 - c. How many weights will be generated in this layer of the model?

1. each kernel position will give 1 output, with a stride of 1, since kernel size is 5x5, means 2 each from the center, thus it gives 8x8x1 output tensor, where 12×12 , $12 - 5 + 1 = 8$ because no padding. Since there are 256 kernels, output is 8x8x256.

2. each kernel postion is 5x5x3, its calculation is a dot product, thus the multiply and accumulate (MAC) operations for each output is 5x5x3, for the size of the kernel, total is 8x8x256, thus MAC total = $5 \times 5 \times 3 \times 8 \times 8 \times 256 = 1228800$ MACs

3. no of weight is equal to number of kernel mulitplied by kernel size, no. of weights in this layer = $256 \times 5 \times 5 \times 3$